# fn score_candidates_map

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March 5, 2026

This proof resides in **"contrib"** because it has not completed the vetting process.

Proves soundness of `score_candidates_map` in `mod.rs` at commit f5bb719 (outdated[1]). `score_candidates_map` returns a specific function that can be used to prove stability of the quantile scoring transformation.

## 1 Hoare Triple

### Precondition

`alpha_den` > `alpha_num`.

### Function

```
def score_candidates_map(alpha_num, alpha_den, known_size) -> Callable[[int], int]:
    def stability_map(d_in):
        if known_size:
            return T.inf_cast(d_in // 2).inf_mul(alpha_den)
        else:
            abs_dist_const: u64 = max(alpha_num, alpha_den - alpha_num)  #
            return T.exact_int_cast(d_in).alerting_mul(abs_dist_const)

    return stability_map
```

### Postcondition

**Theorem 1.1.** Define `function`as follows:

$$\texttt{function}(x) = \text{score_candidates}_i(x, \texttt{candidates}, \texttt{alpha_num}, \texttt{alpha_den}, \texttt{size_limit})$$

The function calls `score_candidates` with fixed choices of `candidates`, `alpha_num`, `alpha_den` and `size_limit`.

If the input domain is the set of all vectors with non-null elements of type `TI`, input metric is either `SymmetricDistance` or `InsertDeleteDistance`, and the postcondition of `function`is satisfied, then `function`is stable.

This means that, for any two elements $x, x'$ in the `input_domain`, where $x, x'$ share the same length if `known_length` is true, and any pair (`d_in`, `d_out`), where `d_in` has the associated type for `input_metric` and `d_out` has the associated type for `output_metric`, then when $x, x'$ are `d_in`-close under `input_metric` and `stability_map`(`d_in`) $\leq$ `d_out`, `function`(x), `function`(x') are `d_out`-close under `output_metric`.

The sensitivity of this function differs depending on whether the size of the input vector is known. First, consider the case where the size is unknown.

[1] See new changes with `git diff f5bb719..34d718e rust/src/transformations/quantile_score_candidates/mod.rs`

**Lemma 1.2.** If $d_{Sym}(x, x') = 1$, then $d_\infty(\texttt{function}(x), \texttt{function}(x')) \leq \max(\alpha_{num}, \alpha_{den} - \alpha_{num})$.

*Proof.* Assume $d_{Sym}(x, x') = 1$.

$$
\begin{aligned}
& d_\infty(\texttt{function}(x)_i, \texttt{function}(x')_i) \\
&= \max_i |\texttt{function}(x)_i - \texttt{function}(x')_i| && \text{by definition of } d_\infty \\
&= \max_i |\text{abs_diff}(\alpha_{den} \cdot \min(\#(x < C_i), l), \alpha_{num} \cdot \min(|x| - \#(x = C_i), l)) && \text{by definition of } \texttt{function} \\
&\qquad \text{abs_diff}(\alpha_{den} \cdot \min(\#(x' < C_i), l), \alpha_{num} \cdot \min(|x'| - \#(x' = C_i), l))| \\
&= \alpha_{den} \cdot \max_i ||\min(\#(x < C_i), l) - \alpha \cdot \min(|x| - \#(x = C_i), l)| \\
&\qquad |\min(\#(x' < C_i), l) - \alpha \cdot \min(|x'| - \#(x' = C_i), l)|| \\
&\leq \alpha_{den} \cdot \max_i ||\#(x < C_i) - \alpha \cdot (|x| - \#(x = C_i))| \\
&\qquad |\#(x' < C_i) - \alpha \cdot (|x| - \#(x' = C_i))||
\end{aligned}
$$

Consider each of the three cases of adding or removing an element in x.

Case 1. Assume $x'$ is equal to $x$, but with some $x_j < C_i$ added or removed.

$$
\begin{aligned}
&= \alpha_{den} \cdot \max_i ||(1-\alpha) \cdot \#(x < C_i) - \alpha \cdot \#(x > C_i)| \\
&\qquad - |(1-\alpha) \cdot (\#(x < C_i) \pm 1) - \alpha \cdot \#(x > C_i)|| \\
&\leq \alpha_{den} \cdot \max_i ||(1-\alpha) \cdot \#(x < C_i) - \alpha \cdot \#(x > C_i)| \\
&\qquad - (|(1-\alpha) \cdot \#(x < C_i) - \alpha \cdot \#(x > C_i)| + |\pm(1-\alpha)|)| && \text{by triangle inequality} \\
&= \alpha_{den} \cdot \max_i |1 - \alpha| && \text{scores cancel} \\
&= \alpha_{den} - \alpha_{num} && \text{since } \alpha \leq 1
\end{aligned}
$$

Case 2. Assume $x'$ is equal to $x$, but with some $x_j > C_i$ added or removed.

$$
\begin{aligned}
&= \alpha_{den} \cdot \max_i ||(1-\alpha) \cdot \#(x < C_i) - \alpha \cdot \#(x > C_i)| \\
&\qquad - |(1-\alpha) \cdot \#(x < C_i) - \alpha \cdot (\#(x > C_i) \pm 1)|| \\
&\leq \alpha_{den} \cdot \max_i ||(1-\alpha) \cdot \#(x < C_i) - \alpha \cdot \#(x > C_i)| \\
&\qquad - (|(1-\alpha) \cdot \#(x < C_i) - \alpha \cdot \#(x > C_i)| + |\pm\alpha|)| && \text{by triangle inequality} \\
&= \alpha_{den} \cdot \max_i |\alpha| && \text{scores cancel} \\
&= \alpha_{num} && \text{since } \alpha \geq 0
\end{aligned}
$$

Case 3. Assume $x'$ is equal to $x$, but with some $x_j = C_i$ added or removed.

$$
\begin{aligned}
&= \alpha_{den} \cdot \max_i ||(1-\alpha) \cdot \#(x < C_i) - \alpha \cdot \#(x > C_i)| \\
&\qquad - |(1-\alpha) \cdot \#(x < C_i) - \alpha \cdot \#(x > C_i)|| \\
&= 0 && \text{no change in score}
\end{aligned}
$$

Take the union bound over all cases.

$$
\leq \max(\alpha_{num}, \alpha_{den} - \alpha_{num})
$$

□

Now consider the case where the dataset size is known.

**Lemma 1.3.** If $d_{CO}(x, x') \leq 1$, then $d_\infty(\texttt{function}(x), \texttt{function}(x')) \leq \alpha_{den}$.

*Proof.* Assume $d_{CO}(x, x') \leq 1$.

$$
\begin{aligned}
& d_\infty(\texttt{function}(x), \texttt{function}(x')) \\
&= \max_i |\texttt{function}(x)_i - \texttt{function}(x')_i| && \text{by definition of } d_\infty \\
&= \max_i |\text{abs_diff}(\alpha_{den} \cdot \min(\#(x < C_i), l), \alpha_{num} \cdot \min(|x| - \#(x = C_i), l)) && \text{by def. of } \texttt{function} \\
&\qquad - \text{abs_diff}(\alpha_{den} \cdot \min(\#(x' < C_i), l), \alpha_{num} \cdot \min(|x'| - \#(x' = C_i), l))| \\
&= \alpha_{den} \cdot \max_i ||\min(\#(x < C_i), l) - \alpha \cdot \min(|x| - \#(x = C_i), l)| \\
&\qquad - |\min(\#(x' < C_i), l) - \alpha \cdot \min(|x'| - \#(x' = C_i), l)|| \\
&= \alpha_{den} \cdot \max_i ||\#(x < C_i) - \alpha \cdot (|x| - \#(x = C_i))| \\
&\qquad - |\#(x' < C_i) - \alpha \cdot (|x| - \#(x' = C_i))||
\end{aligned}
$$

Consider each of the four cases of changing a row in x.

Case 1. Assume $x'$ is equal to $x$, but with some $x_j < C_i$ replaced with $x'_j > C_i$.

$$
\begin{aligned}
&= \alpha_{den} \cdot \max_i ||(1-\alpha) \cdot \#(x < C_i) - \alpha \cdot \#(x > C_i)| \\
&\qquad - |(1-\alpha) \cdot (\#(x < C_i) - 1) - \alpha \cdot (\#(x > C_i) + 1)|| && \text{by definition of } \texttt{function} \\
&\leq \alpha_{den} \cdot \max_i ||(1-\alpha) \cdot \#(x < C_i) - \alpha \cdot \#(x > C_i)| \\
&\qquad - (|(1-\alpha) \cdot \#(x < C_i) - \alpha \cdot \#(x > C_i)| + |1|)| && \text{by triangle inequality} \\
&= \alpha_{den} \cdot \max_i |1| && \text{scores cancel} \\
&= \alpha_{den}
\end{aligned}
$$

Case 2. Assume $x'$ is equal to $x$, but with some $x_j > C_i$ replaced with $x'_j < C_i$.

$$= \alpha_{den}$$

by symmetry, follows from Case 1.

Case 3. Assume $x'$ is equal to $x$, but with some $x_j \neq C_i$ replaced with $C_i$.

$$\leq \max(\alpha_{num}, \alpha_{den} - \alpha_{num})$$

equivalent to one removal (see `make_quantile_score_candidates`)

Case 4. Assume $x'$ is equal to $x$, but with some $x_j = C_i$ replaced with $x'_j \neq C_i$.

$$\leq \max(\alpha_{num}, \alpha_{den} - \alpha_{num})$$

equivalent to one addition (see `make_quantile_score_candidates`)

Take the union bound over all cases.

$$d_\infty(x_i, x'_i) \leq \max(\alpha_{den}, \max(\alpha_{num}, \alpha_{den} - \alpha_{num})) = \alpha_{den}$$

since $\max(\alpha, 1-\alpha) \leq 1$

□

*Proof of postcondition.* Assume the input domain is the set of all vectors with non-null elements of type `TI`, input metric is either `SymmetricDistance` or `InsertDeleteDistance`, and the postcondition of `function`is satisfied.

First, consider the case where the size is unknown. Take any two members $s, s'$ in the `input_domain` and any pair (`d_in`, `d_out`), where `d_in` has the associated type for `input_metric` and `d_out` has the associated type for `output_metric`. Assume $s, s'$ are `d_in`-close under `input_metric` and that `stability_map`(`d_in`) $\leq$ `d_out`.

$$\begin{aligned}
\texttt{d_out} &= \max_{s \sim s'} d_\infty(s, s') && \text{where } s = \texttt{function}(x) \\
&= \max_{s \sim s'} \max_i |s_i - s'_i| && \text{by definition of } \texttt{LInfDistance}\text{, without monotonicity} \\
&\leq \sum_j^{\texttt{d_in}} \max_{Z_j \sim Z_{j+1}} \max_i |s_{i,j} - s_{i,j+1}| && \text{by path property } d_{Sym}(Z_i, Z_{i+1}) = 1, x = Z_0 \text{ and } x' = Z_{\texttt{d_in}} \\
&\leq \sum_j^{\texttt{d_in}} \max(\alpha_{num}, \alpha_{den} - \alpha_{num}) && \text{by 1.2} \\
&\leq \texttt{d_in} \cdot \max(\alpha_{num}, \alpha_{den} - \alpha_{num})
\end{aligned}$$

This formula matches the stability map in the case where the dataset size is unknown.

Now, consider the case where the size is known. Take any two elements $s, s'$ in the `input_domain` and any pair (`d_in`, `d_out`), where `d_in` has the associated type for `input_metric` and `d_out` has the associated type for `output_metric`. Assume $s, s'$ are `d_in`-close under `input_metric` and that `stability_map`(`d_in`) $\leq$ `d_out`.

$$
\begin{aligned}
\texttt{d_out} &= \max_{s \sim s'} d_\infty(s, s') \\
&= \max_{s \sim s'} \max_i |s_i - s'_i| && \text{by definition of } \texttt{LInfDistance}\text{, without monotonicity} \\
&\leq \sum_j^{\texttt{d_in}//2} \max_{Z_j \sim Z_{j+1}} \max_i |s_{i,j} - s_{i,j+1}| && \text{by path property } d_{CO}(Z_i, Z_{i+1}) = 1, x = Z_0 \text{ and } Z_{\texttt{d_in}} = x' \\
&\leq \sum_j^{\texttt{d_in}//2} \alpha_{den} && \text{by 1.3} \\
&\leq (\texttt{d_in}//2) \cdot \alpha_{den}
\end{aligned}
$$

This formula matches the stability map in the case where the dataset size is known.

It is shown that `function`(x), `function`(x') are `d_out`-close under `output_metric` for any choice of input arguments. □